Analysis report

Holzinger–Swineford · ML · listwise · N = 301 · lavaan 0.7.3

Table 1

Parameter estimates

Group	Left	Relation	Right	Label / value	Estimate	SE	z	p	95% CI lower	95% CI upper	std.all
1	visual	=~	x1		1.000	0.000	—	—	1.000	1.000	0.772
1	visual	=~	x2		0.554	0.100	5.554	< .001	0.358	0.749	0.424
1	visual	=~	x3		0.729	0.109	6.685	< .001	0.516	0.943	0.581
1	textual	=~	x4		1.000	0.000	—	—	1.000	1.000	0.852
1	textual	=~	x5		1.113	0.065	17.014	< .001	0.985	1.241	0.855
1	textual	=~	x6		0.926	0.055	16.703	< .001	0.817	1.035	0.838
1	speed	=~	x7		1.000	0.000	—	—	1.000	1.000	0.570
1	speed	=~	x8		1.180	0.165	7.152	< .001	0.857	1.503	0.723
1	speed	=~	x9		1.082	0.151	7.155	< .001	0.785	1.378	0.665
1	x1	~~	x1		0.549	0.114	4.833	< .001	0.326	0.772	0.404
1	x2	~~	x2		1.134	0.102	11.146	< .001	0.934	1.333	0.821
1	x3	~~	x3		0.844	0.091	9.317	< .001	0.667	1.022	0.662
1	x4	~~	x4		0.371	0.048	7.779	< .001	0.278	0.465	0.275
1	x5	~~	x5		0.446	0.058	7.642	< .001	0.332	0.561	0.269

Group	Left	Relation	Right	Label / value	Estimate	SE	z	p	95% CI lower	95% CI upper	std.all
1	x6	~~	x6		0.356	0.043	8.277	< .001	0.272	0.441	0.298
1	x7	~~	x7		0.799	0.081	9.823	< .001	0.640	0.959	0.676
1	x8	~~	x8		0.488	0.074	6.573	< .001	0.342	0.633	0.477
1	x9	~~	x9		0.566	0.071	8.003	< .001	0.427	0.705	0.558
1	visual	~~	visual		0.809	0.145	5.564	< .001	0.524	1.094	1.000
1	textual	~~	textual		0.979	0.112	8.737	< .001	0.760	1.199	1.000
1	speed	~~	speed		0.384	0.086	4.451	< .001	0.215	0.553	1.000
1	visual	~~	textual		0.408	0.074	5.552	< .001	0.264	0.552	0.459
1	visual	~~	speed		0.262	0.056	4.660	< .001	0.152	0.373	0.471
1	textual	~~	speed		0.173	0.049	3.518	< .001	0.077	0.270	0.283

Note. Estimates and intervals are unstandardised; Std. all is fully standardised. Values are rounded only for display. A dash indicates unavailable output. 95% CI.

Table 2

Fit measures

Measure	Value
χ^2	85.306
df	24.000
p (χ^2)	< .001
CFI	0.931
TLI	0.896
RMSEA	0.092
RMSEA 90% CI lower	0.071
RMSEA 90% CI upper	0.114
SRMR	0.065
Scaled χ^2	—
Scaled df	—
p (scaled χ^2)	—
Robust CFI	—
Robust TLI	—
Robust RMSEA	—
Robust RMSEA 90% CI lower	—
Robust RMSEA 90% CI upper	—
Scaled CFI	—
Scaled TLI	—
Scaled RMSEA	—
Scaled RMSEA 90% CI lower	—
Scaled RMSEA 90% CI upper	—

Note. Scaled and robust statistics are labelled explicitly. Compare the appropriate version for your estimator.

Table 3

Explained variance (R^2)

Group	Left	R^2
1	x1	0.596
1	x2	0.179
1	x3	0.338
1	x4	0.725
1	x5	0.731
1	x6	0.702
1	x7	0.324
1	x8	0.523
1	x9	0.442

Table 4

Modification indices

Group	Left	Relation	Right	MI	EPC	Standardised EPC
1	visual	=~	x9	36.411	0.577	0.515
1	x7	~~	x8	34.145	0.536	0.859
1	visual	=~	x7	18.631	-0.422	-0.349
1	x8	~~	x9	14.946	-0.423	-0.805
1	textual	=~	x3	9.151	-0.272	-0.238
1	x2	~~	x7	8.918	-0.183	-0.192
1	textual	=~	x1	8.903	0.350	0.297
1	x2	~~	x3	8.532	0.218	0.223
1	x3	~~	x5	7.858	-0.130	-0.212
1	visual	=~	x5	7.441	-0.210	-0.147
1	x1	~~	x9	7.335	0.138	0.247
1	x4	~~	x6	6.220	-0.235	-0.646
1	x4	~~	x7	5.920	0.098	0.180
1	x1	~~	x7	5.420	-0.129	-0.195
1	x7	~~	x9	5.183	-0.187	-0.278
1	textual	=~	x9	4.796	0.138	0.136
1	visual	=~	x8	4.295	-0.210	-0.187
1	x3	~~	x9	4.126	0.102	0.147
1	x4	~~	x8	3.805	-0.069	-0.162
1	x1	~~	x2	3.606	-0.184	-0.233
1	x1	~~	x4	3.554	0.078	0.173
1	textual	=~	x8	3.359	-0.121	-0.118
1	visual	=~	x6	2.843	0.111	0.092
1	x4	~~	x5	2.534	0.186	0.457

Group	Left	Relation	Right	MI	EPC	Standardised EPC
1	x2	~~	x9	1.895	0.075	0.094
1	x3	~~	x6	1.855	0.055	0.100
1	speed	=~	x2	1.580	-0.198	-0.105
1	x5	~~	x7	1.233	-0.049	-0.083
1	visual	=~	x4	1.211	0.077	0.059
1	x5	~~	x9	0.999	0.040	0.079
1	x1	~~	x3	0.935	-0.139	-0.203
1	x5	~~	x6	0.916	0.101	0.253
1	x2	~~	x6	0.785	0.039	0.062
1	speed	=~	x3	0.716	0.136	0.075
1	x3	~~	x7	0.638	-0.044	-0.054
1	x1	~~	x8	0.634	-0.041	-0.079
1	x2	~~	x4	0.534	-0.034	-0.052
1	x1	~~	x5	0.522	-0.033	-0.067
1	x5	~~	x8	0.347	0.023	0.049
1	x6	~~	x8	0.275	0.018	0.043
1	speed	=~	x6	0.273	0.044	0.025
1	x6	~~	x7	0.259	-0.020	-0.037
1	speed	=~	x5	0.201	-0.044	-0.021
1	x4	~~	x9	0.196	-0.016	-0.035
1	x3	~~	x4	0.142	-0.016	-0.028
1	textual	=~	x7	0.098	-0.021	-0.019
1	x6	~~	x9	0.097	-0.011	-0.024
1	x3	~~	x8	0.059	-0.012	-0.019
1	x2	~~	x8	0.054	-0.012	-0.017

Group	Left	Relation	Right	MI	EPC	Standardised EPC
1	x1	~~	x6	0.048	0.009	0.020
1	x2	~~	x5	0.023	-0.008	-0.011
1	textual	=~	x2	0.017	-0.011	-0.010
1	speed	=~	x1	0.014	0.024	0.013
1	speed	=~	x4	0.003	-0.005	-0.003

Note. Modification indices concern fixed-to-zero parameters, not equality constraints. Add a theoretically justified parameter, then rerun the model. An addition applies only to the indicated group.

Table 5

Reliability & validity

Group	Factor	Composite ω	AVE
1	visual	0.612	0.371
1	textual	0.885	0.721
1	speed	0.686	0.424

Note. Unit-weight omega-type composite reliability and AVE use observed variances (semTools). HTMT2 uses absolute correlations and geometric means; ordinal indicators use polychoric correlations. Reverse-code items in the source data when appropriate. semTools 0.5.10.

Table 6

Discriminant validity

Group	Factor 1	Factor 2	HTMT2	Latent r	$\sqrt{AVE_1}$	$\sqrt{AVE_2}$	Fornell–Larcker
1	visual	textual	0.384	0.459	0.609	0.849	Yes
1	visual	speed	0.387	0.471	0.609	0.652	Yes
1	textual	speed	0.280	0.283	0.849	0.652	Yes

Note. Fornell–Larcker compares each factor's square root of AVE with the absolute fitted latent correlation. “Yes” means both square roots exceed that correlation; it is not a general validity verdict.